

in existing literature [e.g., 21, 23, 42], although in our study there was less emphasis on content authenticity than in prior work.

Worth noting is that while writer participants' definitions of authenticity shared some similarities with existing conceptions of authenticity used by prior literature, they also highlighted additional nuances that diverge from the existing theoretical frameworks of authenticity. Considering the idea of *Source* in authenticity: prior literature took a more retrospective view, asking whether one could attribute a piece of work to its creator(s) (e.g., [4, 21, 42]). Instead, our writer participants focused on various moments during content production: who was taking action and contributing during the actual process of writing? This resembles a more direct way to determine the source of work and its authenticity.

Consider also the concept of *Category* versus *Content authenticity*: while writers in our study also evaluated authenticity in terms of whether the voices and styles of a piece of work represented those of a certain creator(s), they emphasized the professional and economic motivations behind this approach. Many writers described one's unique writing styles as their "fingerprints" and "brand logo," enabling their work to be recognized by broader audiences and their clients. Conceptually, existing frameworks of authenticity assess work by the degree to which it is representative of a certain literature genre or school of thought (e.g., whether a novel captures the essence of postmodernism) [51]. However, our participants rarely judged the authenticity of work by genre; they focused solely on whether a piece of text resembles a writer's signature piece.

Moreover, the writers in our study also centered the concept of authenticity around writers' internal experiences before and throughout the writing process. Specifically, they highlighted the importance of absorbing, digesting, and internalizing information, knowledge, and lived experiences to nourish their authentic selves and express them in writing. Ultimately, this becomes their signature, ensuring that a piece of work could not take its eventual shape if done by anyone else.

This last definition of authenticity raised by our writer participants diverges from that of *Value Alignment* in existing theoretical frameworks defining authenticity. Here, writer participants focused on whether they could genuinely project their internal, living experiences into their writing. In other words, value alignment implies whether participants could represent their genuine selves and lives in their writing. This concept of "value" is distinct from that in the existing literature, which more often refers to writers' ideology, value systems, and ethics [43, 51]. Furthermore, writer participants emphasized that while everyone possesses their own perspectives, emotions, and memories, skills, practices, and iterative processes are required to transform them into writing forms.

Based on these fundamental concepts, we next discuss how the design of human-AI co-writing tools can support writers' authentic selves, source authenticity, and content authenticity.

5.2 Designing AI-writing Tools that Foster *Authenticity* through Personalization

Our findings call attention to how AI writing assistance tools could be designed to support both the different conceptions of authenticity that writers might have and, thus, the *practices* they believe will foster and preserve rather than threaten their sense of authenticity. Specifically, our study reveals that authenticity is multifaceted, and the current form of AI assistance (i.e., text suggestion for sentence completion) only contributes to preserving writers' unique styles and voices in writing. Besides text production, writers seek more diverse support (such as practicing externalizing their internal experiences, receiving feedback, and projecting possible audiences' reactions) to jointly preserve authenticity in their work.

5.2.1 Supporting writers to express their authentic selves through personalized recommendations for sources of writing inspiration. Our study suggests that authenticity could be supported by enriching the materials that writers learn and draw inspiration from and later on apply to (improve) their writing. In our interviews, writers highlighted that the most important practice of “writing” is, in fact, what they do to nurture themselves with rich information and experiences. Many writers emphasized how reading plays an important role in improving their writing, and how AI might be able to efficiently support the process of synthesizing and internalizing large amounts of information. Thus, providing personalized recommendations for what materials the writers might enjoy or need to engage with can support authenticity by helping writers to enrich their authentic, intellectual selves—which are *the* bases for authentic writing.

5.2.2 Providing diverse, early-stage support to help writers establish and accomplish their goals for writing. Across all conversations with writers, we saw how their uncertainty—and oftentimes apprehension—about audiences’ possible responses to their work constrained how freely they felt they could express themselves in writing. Such concerns also explain some of their hesitations to disclose their use of AI when writing, echoing the AI Ghostwriter Effect [13]. However, our survey in Part 2 suggests writers’ worries about how they and their work would be perceived might not necessarily align with readers’ actual attitudes and perceptions of AI-supported writing. To address writers’ concerns about *Source* authenticity (i.e., whether readers would attribute the credit of a piece of work to its author), we first recognize the need to better reflect readers’ opinions about credit and authorship to writers.

Additionally, we see opportunities to build tools that help writers shape their own writing goals and provide feedback accordingly. For example, personalized AI feedback can help writers decide what types of writer identities to present to their readers and whether they want to highlight their use of new technologies to support their work production processes. In particular, offering support during the early stage of idea development (i.e., the “fuzzy area” that writer participants referred to) might be more effective in helping writers preserve their authenticity in writing compared to providing support later on during text production. It is during this early planning stage that writers often decide what goes into the writing, performing “content gatekeeping” that primarily contributes to authenticity, according to many of our writer participants.

5.2.3 Supporting content authenticity through personalizing active feedback. As described in the idea of building a “feedback loop of good writing,” writers in our study were less interested in directly receiving high-quality AI-generated text. Instead, they were more motivated to learn how to improve and produce good writing through AI tools’ feedback. From our interviews, we also learned that what each writer targets and considers “good writing” varies widely from writer to writer. Therefore, providing customized aids to help writers learn to improve their own writing in the ways they want could be another promising use of personalization. In our study, writers specifically sought “machine points of view” for active monitoring and in-text feedback from the AI tool to improve their writing. In this regard, enabling writers to select and customize the types of metrics to use for analyzing their writing output might be a possible starting point.

All in all, we argue that designing AI co-creation tools that preserve authenticity should take a more holistic approach, beginning with the perspective that AI assistance need not be narrowly focused on the writing process and output per se. Instead, designers and developers of these tools could ***target the growth of writers as their ultimate design goal***. This can be done through AI recommending more diverse reading materials to inspire writers and presenting public feedback for writers to retrospect their own work. As writers grow, they may seek different topics, content, and styles in their writing, and they may need support not only in improving the quality of their work but also for them to thrive

intellectually. Therefore, instead of personalizing AI text suggestions, providing personalized assistance and feedback that help writers grow can offer more holistic support in the long run.

5.3 Open Research Questions for Designing AI Writing Assistance

Besides the more concrete design recommendations, as we elaborated in the previous section, writers articulated additional desires and challenges that require future work to explore further the potential of personalized AI tools for addressing these needs. Below, we list four such areas:

Distribution of work: Given their concerns about AI’s direct threats to source authenticity, writers in our study frequently expressed the perspective that AI writing assistance should not be used to automate text production fully. In our study, professional writers still desired to conduct most of their writing work. Many of them acknowledged performing the writing task themselves helped them improve their work quality and grow as writers. However, how to effectively distribute the type and load of work between human writers and AI tools—and thus the appropriate level of AI intervention—remains an open question.

Pacing and temporal adjustment: Throughout our interviews, writer participants described several stages of their writing processes and indicated different types of support needed in each phase. While writers overall desired control over their writing experiences, we noticed the degree of control needed might also differ from stage to stage. In this regard, how to adjust the types and frequency of AI support accordingly is a challenging yet important topic for future work. In particular, personalizing AI assistance through learning individual writers’ patterns might be a potential direction for further exploration.

Communication of writers’ internal states: Reflecting on writers’ living experiences and their internal states remains an under-addressed feature in the design of existing AI writing tools. Our participants not only suggested these elements as the core of authenticity in human writing, but they could also be used as rich materials for composition. In this regard, besides designing tools that can support writers during the writing process, additional tools that can help writers to reflect and express their living experiences *before* writing are likewise useful forms of writing assistance.

Writer-reader relationships in the age of human-AI co-creation: Taking the findings from Part 1 and Part 2 of our study together, we observe differences between writers’ and readers’ views. Despite writers’ concerns, readers could not tell when writers adopted AI assistance and reacted rather positively to the idea of writers’ experimenting with AI co-writing. This suggests a possible difference between how writers and readers judge the *Value* of writing output. For writers, value may be rooted in the hard work that goes into conceiving and producing content; for readers, value may be based on the outcomes of that work, and they do not appear to associate AI-assisted work with a lack of effort. Therefore, devaluation of work may be more likely to arise when writers fail to deliver quality work that meets readers’ expectations—which may not require promises about *how* the work is made.

5.4 What AI Writing Assistance is (Not): Applicability of the Present Insights

Providing more clarity to how authenticity is conceptualized in human-AI co-writing also calls us to reconsider how AI-mediated communication may differ from other AI-assisted writing settings, including the genres we explored in this study. In AIMC, authors adopt AI assistance to produce content for *two-way* communication [22]. This entails delivering messages and meanings to, and expecting reciprocal interaction from, target recipients. Here, the term “mediation” reflects the context where the writing happens, implying AI assistance plays a direct role in altering communication and relationships between writers and readers.